

Architecture and Data Model Specification

Minimum reference architecture, entity model, and control requirements for auditable grid-asset evidence

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OGARD Architecture and Data Model Specification

Version 0.1.2

Deniz Bektas authored this specification using original work, synthetic examples, and public sources.

Companion papers: OGARD Technical Whitepaper; OGARD Implementation Guide; OGARD Synthetic Benchmark Methodology and Validation Plan.

Versioning: This paper is version 0.1.2. The methodology, model profile, and contradiction taxonomy it describes are version 0.1. The executable reference software has its own release numbers (Executable Reference Release v0.2.1 and later).

Repository: <https://github.com/DenBek/ogard>

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Reference implementation license: Apache License 2.0 for original repository code; see LICENSE.md.

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Abstract

This specification defines the minimum architecture and data model needed to implement OGARD as an auditable grid-asset evidence framework. It describes the system boundary, four logical data layers, core entities, temporal identity relationships, mapping and review records, contradiction handling, operator-governed derived asset status, security boundaries, and minimum conformance requirements. The design separates functional location, physical equipment identity, and telemetry identity; preserves source evidence and provenance; routes unresolved mappings to review; and prevents optional artificial-intelligence components from becoming the authority for canonical records or asset status. The specification defines minimum reference controls. Vendor selection, enterprise-model coverage, and deployment design are determined for each operator environment.

Contents

1. Purpose and scope.....	1
2. Minimum design principles and conformance.....	1
3. System context and trust boundaries.....	2
4. Logical architecture.....	3
5. Canonical asset-and-evidence model.....	5
6. Identifier regimes and temporal relationships.....	7
7. Mapping, review, contradiction, and derived status.....	7
8. Interfaces, security, and operational boundaries.....	9
9. Worked synthetic example.....	10
10. Reference repository artifacts and limitations.....	11
References.....	12
Appendix A. Minimum entity dictionary.....	13
Appendix B. Minimum conformance self-assessment.....	14
Appendix C. OGARD v0.1 contradiction taxonomy.....	15
Appendix D. Indicative CIM concept crosswalk.....	16

1. Purpose and scope

The purpose of this document is to make the OGARD architecture, introduced in the OGARD Technical Whitepaper [1], precise enough for technical inspection and reference implementation without turning the framework into an unnecessarily broad utility enterprise standard. It defines the minimum components and records required to preserve evidence, reconcile identity, record uncertainty, and produce controlled information products.

Application scope. This specification defines a reference architecture and data model. Production deployment, operational control, formal Common Information Model (CIM) conformance, and assurance require operator-specific designs, approvals, and evaluation.

1.1. In scope

- Approved source records concerning electricity-network assets.
- Functional location, physical equipment identity, and telemetry identity.
- Evidence preservation, data conformance, identity reconciliation, contradiction handling, human review, and audit.
- Operator-governed derived asset status and diagnostic-support information.
- Public synthetic reference artifacts and operator-controlled implementation boundaries.

1.2. Out of scope

- Operational control commands, protection functions, switching instructions, or direct control of physical assets.
- Autonomous fault diagnosis, failure prediction, or replacement of operator engineering judgment.
- A complete IEC CIM profile, formal conformance claim, or universal utility data model.
- Live utility performance, outage reduction, production reliability, or operator adoption claims.

1.3. Conventions

MUST identifies a minimum requirement for an implementation that claims alignment with this reference specification. SHOULD identifies a recommended control that may be adapted with documented rationale. MAY identifies an optional capability.

2. Minimum design principles and conformance

An implementation does not need to use a particular cloud platform, database, programming language, or deployment topology. It does need to preserve the following controls if it is described as aligned with OGARD.

Table 1. Minimum control profile.

Requirement	Minimum condition
Evidence preservation	Source evidence remains traceable to its original system, identifier, timestamps, schema, and retained payload or payload reference.
Distinct identity regimes	Functional location, physical equipment identity, and telemetry identity are represented separately and related through time-valid records.
Temporal relationships	Installation and identity relationships carry valid-from and valid-to dates or an equivalent versioned history.
Auditable reconciliation	Each accepted, rejected, deferred, or unresolved mapping has a recorded method, rule or score, version, outcome, and actor where applicable.
Explicit uncertainty	Insufficient or conflicting evidence may result in review, contradiction, or Unknown status rather than a forced mapping or status.
Operator authority	Only authorized rules and reviewers may approve canonical relationships or derived asset status.
Contradiction preservation	Material disagreement is recorded and routed; it is not silently overwritten by survivorship logic.
Model traceability	Persisted material records identify or trace to model_profile_version and schema_version.
Protected-data boundary	Public artifacts contain no employer, client, or operator-controlled sensitive data.
AI limitation	Any AI capability is advisory, logged, evidence-bound, and unable to mutate canonical state or determine final asset status directly.

Conformance scope. Alignment means satisfying the minimum control profile above. Formal standards conformance and certification require their respective assessment processes.

2.1. Conformance self-assessment

An implementation claiming alignment should complete the objective pass/fail checklist in Appendix B and retain the cited evidence. A failed control means the implementation is not aligned for that release scope unless the claim is narrowed to exclude the unmet capability. Self-assessment is not certification; formal certification follows the applicable external assessment process.

3. System context and trust boundaries

OGARD has two deliberately separate contexts. Public artifacts explain and demonstrate the method using original, synthetic, or public data. Any operator implementation runs inside an operator-controlled environment and inherits that operator's security, governance, retention, access, and review requirements.

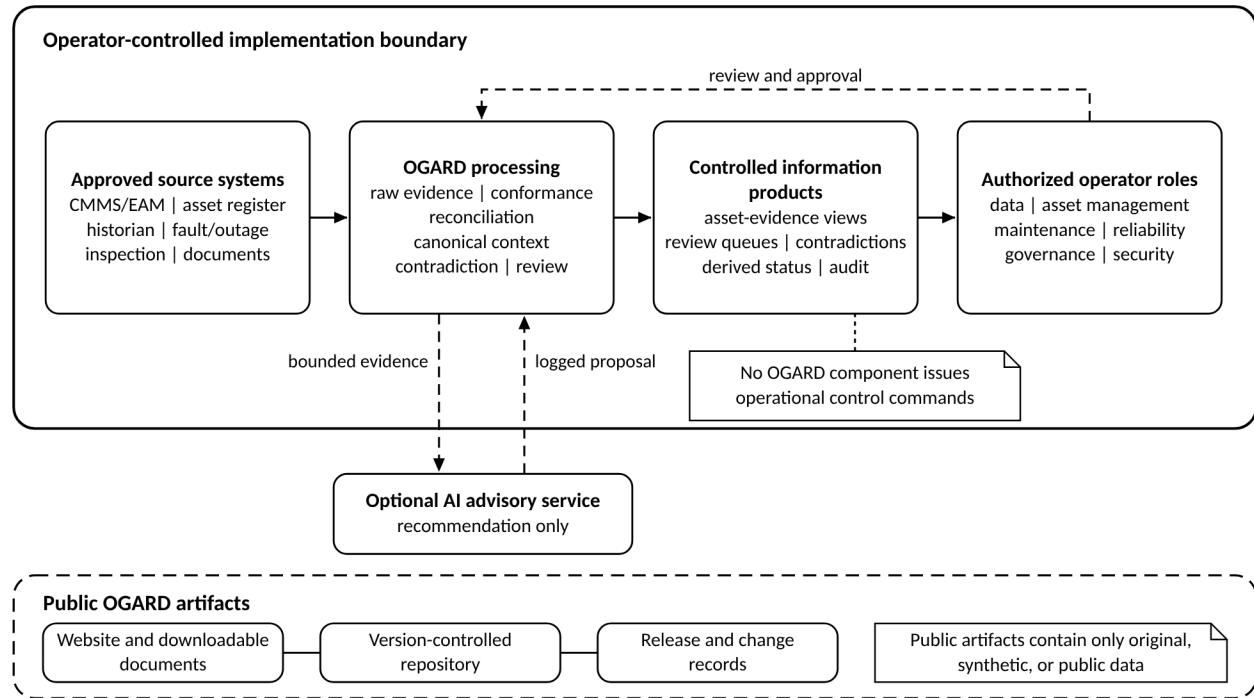


Figure 1. System context and trust boundaries.

3.1. Public reference boundary

The public website and repository MAY contain documentation, diagrams, schemas, synthetic examples, rules, evaluation logic, expected outputs, and review records. They MUST NOT contain protected operator records, credentials, internal configurations, employer code, client information, or proprietary product material.

3.2. Operator-controlled boundary

An operator implementation MUST define source owners, data-access permissions, security classification, retention, reviewer roles, decision rights, and approved downstream use before controlled outputs are relied upon. OGARD does not authorize data access or operational use by itself. The OGARD Implementation Guide [2] sets out the corresponding implementation sequence and controls.

3.3. No control-plane function

OGARD outputs are information products. The reference architecture MUST NOT issue switching commands, protection actions, control-room instructions, or direct changes to operational technology. Any downstream operational action remains governed by the operator's approved systems and procedures.

4. Logical architecture

The reference architecture uses four logical layers. The layers may be implemented as separate services, schemas, tables, or processing stages, but their responsibilities and audit boundaries should remain distinguishable.

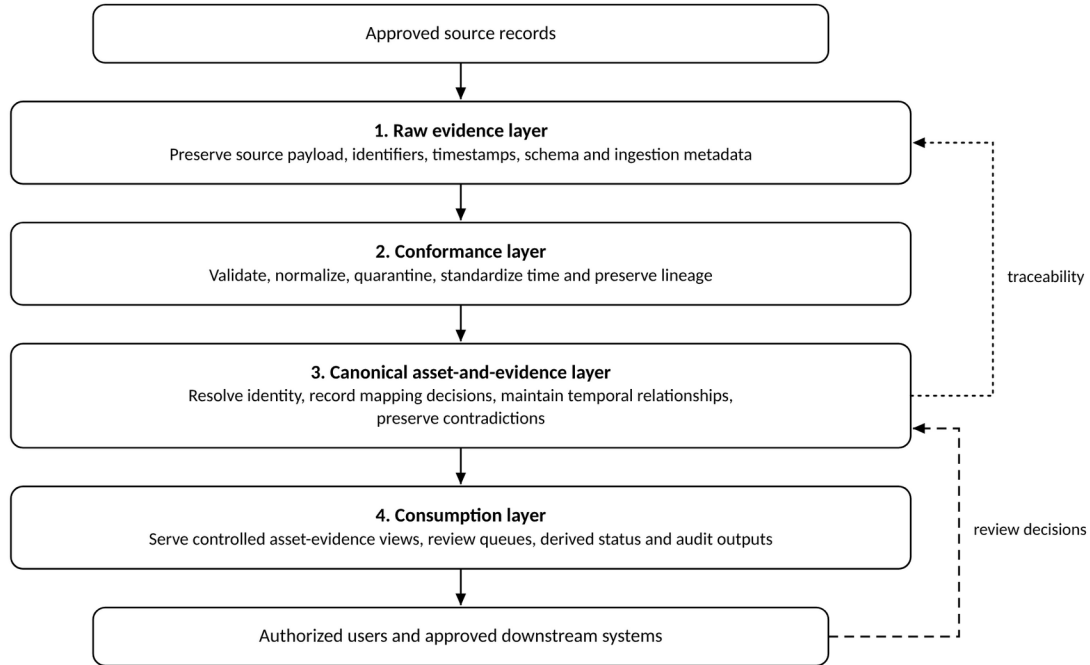


Figure 2. Minimum logical architecture.

Table 2. Logical layers and minimum guarantees.

Layer	Primary responsibility	Minimum guarantee
Raw evidence	Capture or reference source records with source identifiers, original values, timestamps, schema, and ingestion metadata.	The source record can be recovered or traced without relying on the canonical view.
Conformance	Validate schemas, normalize permitted fields, standardize time and units, identify duplicates, and route failures.	Normalization and quality decisions are explicit and do not erase the raw evidence.
Canonical asset-and-evidence	Maintain asset context, time-valid relationships, mapping decisions, review outcomes, contradictions, and derived status provenance.	Every material relationship or status can be explained from evidence, rules, and review records.
Consumption	Provide controlled asset-evidence views, queues, audit outputs, and approved information products.	Consumers receive authorized outputs with lineage and known quality or review state.

4.1. Source contract

Every ingested or referenced source record **MUST** provide, directly or through a controlled catalog, the source system, source record identifier, record type, event timestamp where available, extraction timestamp, ingestion timestamp, schema version, batch or stream identifier, and security classification. Missing required metadata **MUST** be recorded as a quality condition.

4.2. Failure behavior

Records that fail critical validation **SHOULD** be quarantined or excluded from controlled downstream use without being silently deleted. Processing failures, schema drift, stale sources, and incomplete audit records **MUST** be observable.

5. Canonical asset-and-evidence model

The model separates the asset context from the evidence records that describe or challenge it. The canonical view is therefore not a replacement for source records. It is a governed interpretation whose basis remains visible.

Versioning requirement. Every persisted canonical, evidence, mapping, review, contradiction, derived-status, and audit record **MUST** carry, or be traceable to, both `model_profile_version` and `schema_version`. Historical outputs **MUST** remain traceable to the model and schema in force when the decision was produced; later model changes **MUST NOT** silently reinterpret prior records.

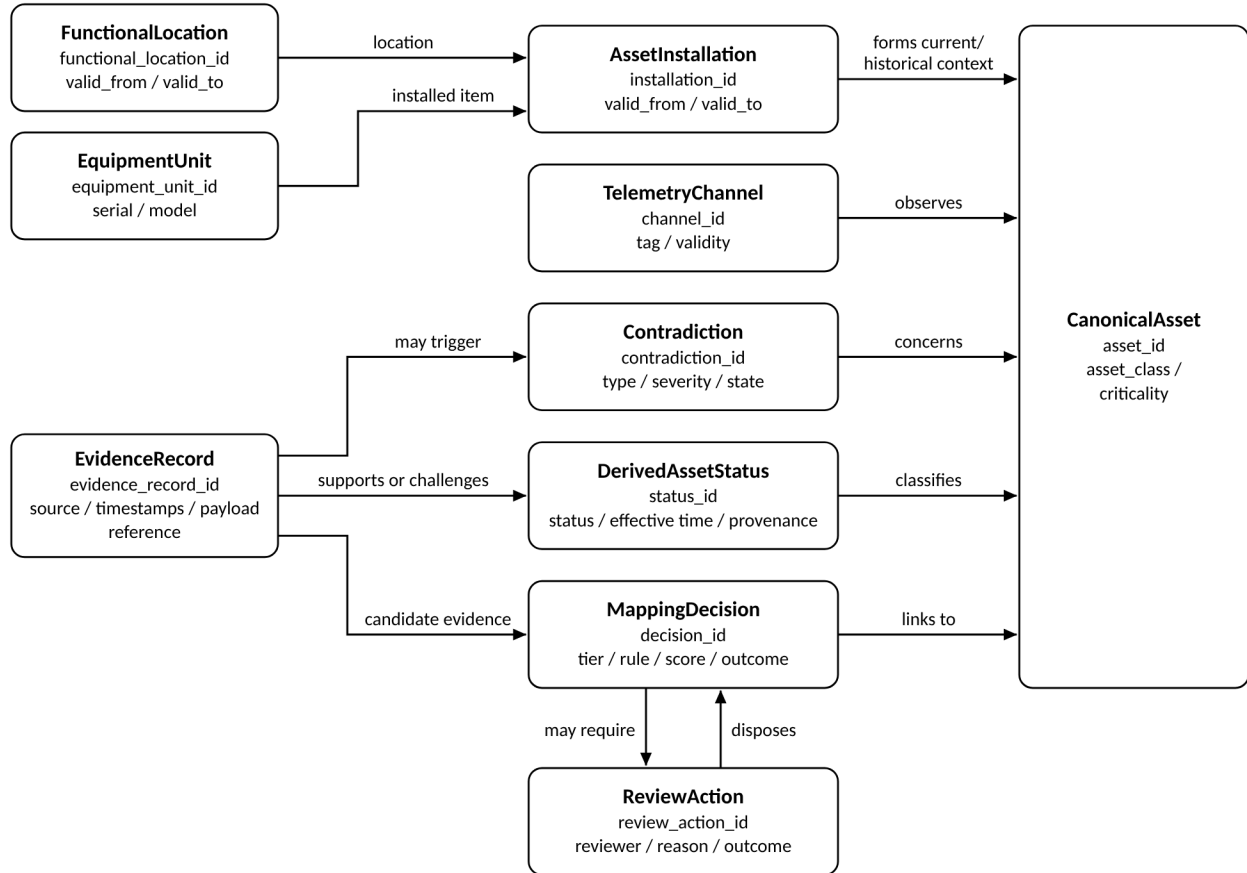


Figure 3. Core entity relationships.

Table 3. Core entities and minimum information.

Entity	Purpose	Minimum information
CanonicalAsset	Stable governed reference context for an asset within the selected implementation scope.	asset_id, asset_class, operator scope, created_at, updated_at; lifecycle or criticality fields only where defined; model_profile_version; schema_version
FunctionalLocation	Role or position in the network hierarchy.	functional_location_id, hierarchy or parent, location type, valid_from, valid_to, model_profile_version; schema_version
EquipmentUnit	Physical installed item.	equipment_unit_id, serial or alternative identity, equipment class, manufacturer/model where available, lifecycle dates, model_profile_version; schema_version
AssetInstallation	Time-valid relationship between a functional location and equipment unit.	installation_id, functional_location_id, equipment_unit_id, valid_from, valid_to, evidence reference, model_profile_version; schema_version
TelemetryChannel	Observation channel associated with a location, item, or aggregate.	channel_id, source system, tag, measurement type, unit, valid_from, valid_to, mapping state, model_profile_version; schema_version
EvidenceRecord	Preserved source-level record or controlled source reference.	evidence_record_id, source system, source record ID, record type, timestamps, batch or stream identifier, security classification, schema version, payload reference, quality state, model_profile_version
EvidenceRecord subtype examples	Typed extensions for maintenance or work orders, faults or outages, inspections or condition records, and technical documents.	evidence_record_id, record_type, subtype-specific fields, source references, model_profile_version; schema_version
MappingDecision	Auditable disposition of a candidate relationship.	decision_id, source record, candidate context, method tier, rule or model identifier and version, score or confidence, threshold configuration, outcome, actor, reason code, timestamps, effective validity period, evidence references, model_profile_version; schema_version
ReviewAction	Authorized human disposition of a review item.	review_action_id, review item, reviewer, decision, reason code, timestamp, evidence references, model_profile_version; schema_version
Contradiction	Typed disagreement or unresolved conflict between evidence classes.	contradiction_id, type, severity, implicated context, triggering evidence, state, owner, resolution, model_profile_version; schema_version
DerivedAssetStatus	Operator-governed status with evidence, rule, time, and review provenance.	status_id, asset_id, status, effective interval, evidence set, rule version, review state, contradiction references, model_profile_version; schema_version

WorkOrder, FaultEvent, InspectionRecord, and DocumentReference are treated as typed EvidenceRecord subtypes or implementation extensions. They do not replace the core EvidenceRecord and provenance requirements.

5.1. Relationship cardinality and time

A functional location MAY be occupied by different equipment units over time. An equipment unit MAY move between locations where the operator permits such history. A telemetry channel MAY observe a functional location, an installed item, or an aggregate and MUST therefore have its own time-valid mapping. Historical relationships SHOULD be closed with valid_to rather than overwritten.

5.2. Selected CIM relationship

OGARD MAY map entities to selected IEC Common Information Model concepts [4], [5] to improve interoperability [3]. The mapping should be documented as guidance unless a version-specific profile, serialization, and conformance process have been implemented. Appendix D provides a short indicative crosswalk that supports the term CIM-informed without claiming a formal profile, complete semantic equivalence, compliance, or certification.

6. Identifier regimes and temporal relationships

Table 4. Identifier regimes.

Identifier regime	Represents	Common failure mode
Functional location	The role or position in the network hierarchy.	It survives equipment replacement and may be mistaken for the physical item.
Physical equipment identity	The installed item, commonly represented by serial, manufacturer, or operator identifier.	It may relocate, be replaced, or lack complete historical identity.
Telemetry identity	The tag or channel through which behavior is observed.	Tags may be renamed, reused, repointed, or left stale after configuration changes.

The model MUST NOT assume that equality within one identifier regime establishes equality across the others. Candidate mappings SHOULD use time windows, asset class, location context, approved aliases, replacement history, and source authority where available. Ambiguous relationships MUST be routed to review or remain unresolved.

7. Mapping, review, contradiction, and derived status

7.1. Reconciliation workflow

1. Generate a bounded candidate set using location, asset class, identifier fragments, time windows, approved aliases, or other documented constraints.
2. Apply versioned deterministic rules where the relationship is sufficiently explicit.
3. Apply reproducible similarity scoring only where deterministic evidence is incomplete and the method is approved for the source and asset class.
4. Route ambiguous, contradictory, or operationally consequential cases to an authorized review queue.
5. Record the final outcome, evidence, method, version, actor, and timestamps in the mapping and audit records.

The method_tier field records the method applied: Tier 1, versioned deterministic rules (step 2); Tier 2, reproducible similarity scoring (step 3); Tier 3, optional machine-assisted recommendation, advisory only (Section 8.3); or authorized human review (step 4).

Table 5. Decision outcomes.

Decision outcome	Meaning
Accepted	The relationship is approved for the stated scope and validity period.
Rejected	The candidate relationship is not approved and the reason is recorded.
Deferred or unresolved	Evidence is insufficient; the case remains unresolved pending additional information.
Escalated	A specialist or higher-authority reviewer is required.
Contradiction recorded	The disagreement is preserved and routed without forcing a single answer.
Routed to review	Interim state awaiting an authorized review decision; not a final disposition.

7.2. Minimum mapping decision record

Table 6. Minimum mapping decision record.

Field group	Required content
Identity	decision_id, source_record_id, candidate_asset_id or candidate set
Method	method tier, rule or model identifier, version, score or confidence where applicable, threshold configuration
Outcome	accepted, rejected, deferred or unresolved, escalated, contradiction recorded, or routed to review (interim)
Authority	automatic rule authority or reviewer identity and reason code
Time	decision timestamp, effective validity period where applicable
Evidence	source evidence references and any competing candidates or contradictions
Versioning	model_profile_version and schema_version

7.3. Contradiction handling

A contradiction **MUST** be preserved where material evidence disagrees on identity, time, status, installed configuration, telemetry association, or maintenance context. Contradictions **MUST** be typed with a versioned code and **SHOULD** be assigned severity under operator-approved rules, routed to an owner, and closed only through a recorded resolution or accepted residual risk. Appendix C defines the closed, versioned OGARD v0.1 contradiction taxonomy used for any benchmark precision or recall calculation. Operator implementations **MAY** extend the taxonomy, but extensions **MUST** use separate versioned codes and **MUST NOT** be mixed into v0.1 benchmark denominators without disclosure.

7.4. Derived asset status

Table 7. Derived asset status.

Status	Reference meaning	Minimum interpretation safeguard
Available	Approved evidence supports expected availability within the defined scope, and no unresolved material contradiction blocks the classification.	Availability is assessed within the defined evidence scope. The operator defines coverage, freshness, and disqualifying contradictions; latent defects require separate assessment.
Degraded	Approved evidence supports impaired condition or reduced capability under an operator-defined rule.	Requires asset-class-specific criteria, time validity, and authorized engineering interpretation.
Offline	Approved evidence supports that the asset is unavailable, disconnected, or out of service.	Must distinguish planned, forced, temporary, and data-quality conditions using time-valid evidence.
At risk	Approved evidence indicates a condition requiring attention but does not support Degraded or Offline.	Must not be represented as a failure prediction; indicators, expiry, and review requirements must be defined.
Unknown	Evidence is insufficient, stale, ambiguous, outside scope, or materially contradictory.	Unknown is the safe default whenever another status cannot be justified.

Status interpretation. Use Unknown when the available evidence is insufficient to support another status, preserving the uncertainty for review.

Every derived status **MUST** record the applicable rule version, evidence set, effective time, review state, and any unresolved contradiction that affects interpretation. AI outputs **MUST NOT** determine final status directly.

8. Interfaces, security, and operational boundaries

8.1. Minimum interface records

Reference interfaces may be implemented as files, tables, messages, APIs, or controlled views. Regardless of technology, the implementation **SHOULD** expose stable records for source evidence, mapping decisions, review actions, contradictions, derived status, and audit events. Schema changes **MUST** be versioned and assessed for backward-compatibility impact.

8.2. Security and confidentiality

- Access **MUST** follow operator-approved least-privilege roles and environment separation.
- Credentials, secrets, and restricted payloads **MUST NOT** be stored in public artifacts.
- Sensitive data **SHOULD** be encrypted in transit and at rest according to operator policy.
- Lineage, decision records, and security-relevant access **SHOULD** be auditable.
- Retention and deletion **MUST** follow applicable legal, contractual, regulatory, and operational requirements.

8.3. Optional AI boundary

An optional AI component **MAY** recommend candidate mappings, summarize evidence, classify documents, or prioritize review. Each use **MUST** identify the task, model and version where available, permitted inputs, output schema, evidence references, confidence or uncertainty, review requirement, retention, and

evaluation method. The component **MUST NOT** write canonical relationships or final asset status without an approved non-AI authority path. An AI service outside the operator-controlled boundary **MAY** receive only operator-approved, bounded evidence under the controls in Section 8.2.

8.4. Observability

At minimum, an implementation **SHOULD** monitor source freshness, schema failures, quarantine volume, mapping outcomes by tier, review-queue depth and age, unresolved contradictions, status freshness, audit-log completeness, and controlled-output availability.

9. Worked synthetic example

A synthetic maintenance record references functional location FL-NE-SUB114-TX02 and contains no equipment serial number. Installation history shows former equipment unit EQ-TR-OLD-01 (serial TR-88-104553) at the location until 14 April 2026 and replacement unit EQ-TR-NEW-01 (serial TR-26-778204) from 15 April 2026. Historian tag NE114.TX2.OILTMP remains explicitly associated with the former equipment unit after replacement. The resulting stale tag-to-equipment relationship is preserved as a contradiction.

1. The functional-location rule links the work order to the location, but not automatically to a physical equipment unit.
2. The installation table limits candidate equipment units using the work-order date.
3. An illustrative similarity score (0.78) and date context identify the replacement unit as the leading candidate, but the score remains below the operator's auto-accept threshold.
4. The case is routed to review with the candidate set, source evidence, time window, and score.
5. A synthetic authorized reviewer accepts the replacement unit and records the reason and evidence references.
6. The stale telemetry association to the former unit is preserved as contradiction CON-TELEM-STALE-001 (TELEMETRY_EQUIPMENT_CONFLICT, state open), pending resolution by an authorized telemetry or historian owner.
7. Until the contradiction is resolved, any status that depends on that telemetry remains Unknown or carries an explicit limitation under operator rules.

This example demonstrates the intended control pattern: identity is resolved through time-valid evidence, the review decision is auditable, and contradictory information is retained rather than silently overwritten.

These records are published as the non-executable worked example (worked-example/*.json). Executable Reference Release v0.2.1 applies deterministic rules only and routes the same work order to review because it has no equipment identifier.

10. Reference repository artifacts and limitations

10.1. Minimum repository artifacts

Table 8. Minimum repository artifacts.

Artifact	Purpose
Architecture diagrams	Editable or source-controlled versions of the system context, logical architecture, and entity model.
Machine-readable schemas	Reference definitions for the core entities and minimum records.
Example records	Synthetic examples showing valid, invalid, ambiguous, and contradictory cases.
Rule and threshold examples	Versioned starter rules and non-production scoring configuration.
Worked reconciliation	A traceable example from source evidence to final disposition.
Audit-log sample	Example accepted, rejected, reviewed, and contradiction events.
Change log and license	Version history, scope changes, and permitted use.

10.2. Known limitations

- The model is a minimum reference profile, not a complete utility enterprise model.
- Asset-class semantics, source authority, quality thresholds, and derived-status rules require operator and domain input.
- Formal CIM conformance requires a complete profile and conformance testing in addition to the mapping guidance.
- Synthetic examples cannot reproduce the full history, irregularity, access constraints, or organizational context of a live estate.
- Production suitability and operational reliability require evaluation in the intended operator environment.

10.3. Release status

This version is published with a tagged repository release and change history. The diagrams, worked example, and reference files are published; machine-readable schema definitions for the core entities are a development item.

10.4. Executable reference release

Executable Reference Release v0.2.1 implements a deterministic subset of this specification (no similarity scoring or derived status). It records two implementation-specific contradiction codes outside the closed v0.1 taxonomy: `COMPETING_INSTALLATIONS` and `EXPLICIT_EQUIPMENT_CONFLICT`.

References

- [1] Bektas, D. OGARD Technical Whitepaper: Open Grid Asset Reliability Data Framework. Version 0.1.2, 2026. <https://github.com/DenBek/ogard>
- [2] Bektas, D. OGARD Implementation Guide: Implementing an Auditable Grid-Asset Evidence Foundation. Version 0.1.2, 2026. <https://github.com/DenBek/ogard>
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- [4] International Electrotechnical Commission. IEC 61970-301:2020+AMD1:2022, Energy management system application program interface (EMS-API) – Part 301: Common information model (CIM) base. <https://webstore.iec.ch/en/publication/74467>
- [5] International Electrotechnical Commission. IEC 61968-11:2013, Application integration at electric utilities – System interfaces for distribution management – Part 11: Common information model (CIM) extensions for distribution. <https://webstore.iec.ch/en/publication/6199>

Appendix A. Minimum entity dictionary

The following field sets are intentionally concise. A reference implementation may add fields, but it should not remove the provenance, identity, time, or decision information needed to reconstruct a material output.

Table 9. Minimum entity dictionary.

Entity	Minimum fields
CanonicalAsset	asset_id; asset_class; operator_scope; created_at; updated_at; optional lifecycle or criticality fields only where defined; model_profile_version; schema_version
FunctionalLocation	functional_location_id; parent_location_id; location_type; valid_from; valid_to; model_profile_version; schema_version
EquipmentUnit	equipment_unit_id; serial_or_alternative_id; equipment_class; manufacturer/model if available; lifecycle dates; model_profile_version; schema_version
AssetInstallation	installation_id; functional_location_id; equipment_unit_id; valid_from; valid_to; evidence_reference; model_profile_version; schema_version
TelemetryChannel	channel_id; source_system; tag; measurement_type; unit; valid_from; valid_to; mapping_state; model_profile_version; schema_version
EvidenceRecord	evidence_record_id; source_system; source_record_id; record_type; event/extraction/ingestion timestamps; batch_or_stream_id; security_classification; schema_version; payload_reference; quality_state; model_profile_version
EvidenceRecord subtype examples	evidence_record_id; record_type; subtype-specific fields; source references; model_profile_version; schema_version
MappingDecision	decision_id; source_record_id; candidate_asset_id or candidate set; method_tier; rule_or_model_id; version; score or confidence; threshold_configuration; outcome; actor; reason_code; timestamp; effective validity period; evidence_references; model_profile_version; schema_version
ReviewAction	review_action_id; review_item_id; reviewer; decision; reason_code; timestamp; evidence_references; model_profile_version; schema_version
Contradiction	contradiction_id; type; severity; implicated_context; triggering_evidence; state; owner; resolution; model_profile_version; schema_version
DerivedAssetStatus	status_id; asset_id; status; effective_from; effective_to; rule_version; evidence_set; review_state; contradiction_references; model_profile_version; schema_version

Appendix B. Minimum conformance self-assessment

The checklist covers controls demonstrated by records or configuration. The assessor should record Pass, Fail, or Not applicable with the supporting evidence and a documented scope reason.

Table 10. OGARD v0.1 minimum conformance checklist.

Control	Objective pass condition	Minimum evidence
Evidence preservation	Every controlled output traces to a source record or controlled payload reference with source and time metadata.	Lineage walkthrough and sample records
Distinct identity regimes	Functional location, equipment unit, and telemetry channel are represented as separate entities or records.	Schema and example records
Temporal relationships	Installation and identity relationships carry valid-from and valid-to semantics or an equivalent versioned history.	AssetInstallation or equivalent history
Auditable reconciliation	Each accepted, rejected, deferred or unresolved, escalated, or reviewed mapping records method, version, outcome, time, and authority.	MappingDecision and ReviewAction samples
Explicit uncertainty	Insufficient evidence can result in review, deferred disposition, contradiction, or Unknown rather than a forced answer.	Rules and test or example cases
Contradiction preservation	Material disagreement is recorded with a versioned type and is not silently overwritten by survivorship.	Contradiction record and resolution workflow
Operator authority	Final canonical relationships and derived statuses are approved only through authorized rules or reviewers.	Decision-rights register and rule authority
Model traceability	Persisted material records identify or trace to model_profile_version and schema_version.	Record samples and schema bundle
Protected-data boundary	Public artifacts contain only original, synthetic, or authorized public information.	Release review and repository inspection
AI limitation	Any AI function is advisory, logged, evidence-bound, and unable to mutate canonical state or determine final status directly.	Architecture and AI control record, if AI is used

Appendix C. OGARD v0.1 contradiction taxonomy

The following codes form the closed v0.1 benchmark taxonomy. A seeded contradiction **MUST** use exactly one primary code for metric accounting, although secondary context may be retained. Operator-specific extensions **MUST** use a separately versioned namespace and **MUST NOT** be mixed into v0.1 benchmark denominators without disclosure. Where a stale telemetry-to-equipment association conflicts with time-valid installation evidence, **TELEMETRY_EQUIPMENT_CONFLICT** is the primary code; **STALE_EVIDENCE_CONFLICT** applies only when no more specific code applies.

Table 11. Closed and versioned contradiction types for OGARD v0.1.

Code	Definition	Illustrative trigger	Expected safe treatment
STATUS_CONFLICT	Approved evidence sources report incompatible asset states for overlapping time.	Register active while fault or outage evidence reports offline.	Preserve both sources; route status review.
EQUIPMENT_LOCATION_CONFLICT	Equipment identity and functional-location history cannot both be true for the same interval.	One serial appears at two locations during overlapping validity.	Review installation history; do not force identity.
TELEMETRY_EQUIPMENT_CONFLICT	A telemetry channel association conflicts with time-valid equipment-installation evidence.	Tag remains mapped to the former unit after replacement.	Preserve channel evidence; route historian or instrumentation review.
TEMPORAL_VALIDITY_CONFLICT	Records imply relationships or events outside compatible validity intervals.	Late work order references equipment removed before the event.	Use event time; review or reject the relationship.
DUPLICATE_IDENTITY_CONFLICT	Competing records claim the same identity or one record appears to represent multiple assets.	Two active equipment records share one serial.	Retain candidates; review duplicate identity.
DOCUMENTED_CONFIGURATION_CONFLICT	Technical documentation and recorded installed configuration disagree.	Drawing names a different installed model or connection.	Route engineering document and configuration review.
WORK_ORDER_REFERENCE_CONFLICT	A maintenance or work-order reference points to an incompatible asset context.	Work order location and equipment reference disagree.	Review source context and installation history.
STALE_EVIDENCE_CONFLICT	A source remains unchanged beyond an approved freshness or change event and conflicts with newer evidence.	Old register status persists after an approved replacement.	Flag staleness; do not infer physical failure.

Appendix D. Indicative CIM concept crosswalk

This crosswalk is non-normative and intentionally incomplete. It indicates related CIM concepts that may support implementation analysis; it does not assert one-to-one equivalence or a version-specific CIM profile.

Table 12. Indicative, non-normative CIM relationship guidance.

OGARD concept	Indicative CIM concept family	Relationship	Limitation
CanonicalAsset	PowerSystemResource and related asset context	Provides the governed OGARD context to which evidence is attached.	Not asserted as a direct CIM class equivalence.
FunctionalLocation	EquipmentContainer hierarchy (for example Substation, VoltageLevel, Bay)	Represents the enduring network role or position.	CIM Location is geographic placement only; utility functional-location semantics vary.
EquipmentUnit	Asset and asset-information concepts (IEC 61968-11)	Represents the physical installed item and lifecycle identity.	CIM Equipment models the network role, not the physical unit.
AssetInstallation	Asset-PowerSystemResource association with Asset lifecycle dates	Bridges equipment identity and functional location through time.	OGARD-specific relationship assembled from relevant CIM concepts.
TelemetryChannel	Measurement and MeasurementValue-related concepts	Represents the observation channel and its mapping context.	Historian tags and channel lifecycles require local extensions.
EvidenceRecord	No single direct equivalent; may reference CIM messages or domain records	Preserves source evidence, provenance, and quality state.	OGARD evidence wrapper extends beyond a semantic equipment model.
MappingDecision / Contradiction	No single direct equivalent	Records governance decisions and unresolved disagreement.	OGARD control records defined by this reference model.